

Abstract Algebra II - Homework 3

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Theorem 2.1 (Isomorphism extension theorem). *Given any field F , an algebraic extension field E of F and an isomorphism φ mapping F onto a field F' then φ can be extended to an isomorphism τ mapping E onto an algebraic extension E' of F' (a subfield of the algebraic closure of F').*

Theorem 2.2. *Let F be a field of characteristic 0 and let $f(x) \in F[x]$, and let G be the Galois group of $f(x)$ over F . If $f(x)$ is irreducible, G acts transitively on the roots of $f(x)$.*

Theorem 2.3. *Symmetric group S_p (where p is a prime) can be generated by a transposition and a p -cycle.*

Proof. Consider any transposition $\tau = (ab)$, and the p -cycle $\sigma = (12 \dots p)$. Because $\langle \sigma \rangle$ is the subgroup of S_p consists of p -cycles, and every element has order p (p is prime) and can be the generator. Hence, there exists integer k such that σ^k sends a to b and is a p -cycle. $\square$